

Question Paper

Exam Date & Time: 31-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

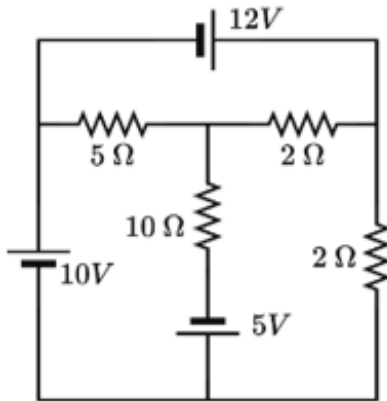
FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025/JAN 2026
SUBJECT: ELE 1003/ELE_1003 FUNDAMENTALS OF ELECTRICAL ENGINEERING

Marks: 50

Duration: 180 mins.

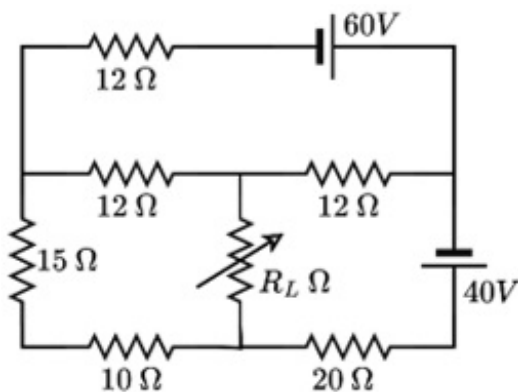
Answer all the questions.

- 1A) For the circuit shown below, determine the power dissipated by the $10\ \Omega$ resistor. (5)



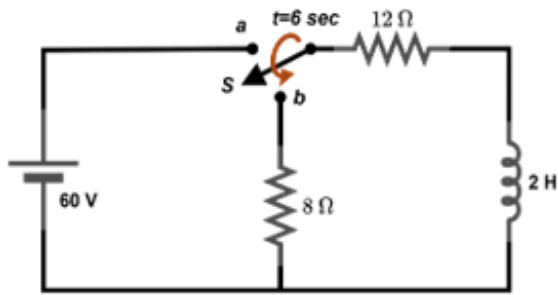
Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 1B) For the given circuit, determine the value of the load resistance R_L , for which maximum power transfer occurs across it. Also, determine the maximum power value. (5)



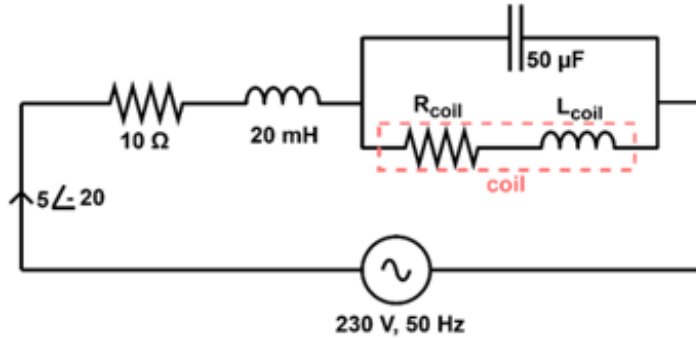
Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 2A) In the circuit shown, the switch is in position 'a' at $t = 0$. After 6 seconds, the switch is moved to position 'b'. (3)
a) Determine the expression for current through the inductor for $t \geq 6$ seconds.
b) Sketch the current response



Course Outcome: "CO1" / Cognitive Level: "Application"

- 2B) In the circuit shown below, determine the resistance and inductance of the coil. (3)



Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 2C) A single-phase 230V, 50Hz source is connected to the following loads, which are connected in parallel. (4)
- a) 2000W, 0.707 lag
 - b) 1500 Var, 0.6 lag
- Find the total current drawn by the loads. Also, determine the value of the capacitance to be connected across the loads, to improve the power factor to 0.95 lagging.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

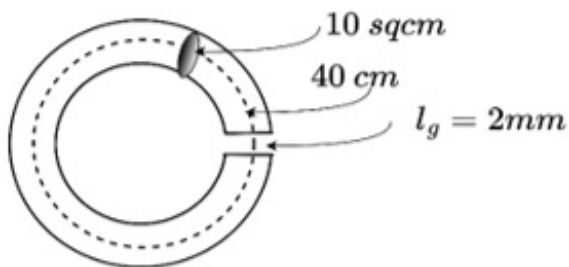
- 3A) A 3-phase, 4-wire, 440V, 50Hz, RYB supply supplies a star-connected load with the phase impedances: $Z_R = Z_Y = Z_B = 6 + j8 \Omega$. Taking V_{RN} as reference, calculate, (6)
- a) Line currents
 - b) Neutral Current
 - c) Total active power
 - d) Draw the phasor diagram

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3B) Each of the two wattmeters connected to measure the input to a 3-phase circuit reads 20 kW. What does each instrument read when the load power factor is 0.866 lagging, with the total 3-phase power remaining unchanged? (4)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 4A) A ring made of ferromagnetic material with a cross-sectional area of 10 cm^2 and a mean circumference of 40 cm is wound uniformly with a coil of 500 turns. Calculate the total reluctance and the current required to produce a flux density of 1.6 T in the air gap of length 2 mm. The relative permeability of the ferromagnetic material is 1000 (4)



Course Outcome: "CO3" / Cognitive Level: "Application"

- 4B) Compare the efficiency of transformers and induction motors. Explain why transformers are generally more efficient. List the different types of losses present in both. (3)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 4C) Describe any two sustainable methods for electric power generation. (3)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 5A) With a neat sketch, explain the working of a three-phase digital energy meter. (3)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 5B) With a neat diagram, explain the layout of an Uninterruptible Power Supply (UPS). Also, list the different types of UPS. (3)

Course Outcome: "CO5" / Cognitive Level: "Understand"

- 5C) Draw the block diagram of a laptop's power stages and describe the functionality of its key components. (4)

Course Outcome: "CO5" / Cognitive Level: "Understand"

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Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	ELE_1003:FUNDAMENTALS OF ELECTRICAL ENGINEERING

#	Course Outcome	Description	Marks
1	1	CO 1	13
2	2	CO 2	17
3	3	CO 3	7
4	4	CO 4	6
5	5	CO 5	7
Total			50

* **NA** - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	ELE_1003:FUNDAMENTALS OF ELECTRICAL ENGINEERING

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	5	16(32%)
3	Application	4	17(34%)

4	Analysis	4	17(34%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	13	50 (100%)